

# ETHAN TRAN

ethantran@quwin.dev · github.com/quwin · linkedin.com/in/kien-ethan-tran · www.quwin.dev

## SUMMARY

Computer Science junior focused on full-stack engineering and distributed services. Experienced building production-style APIs, web platforms, data pipelines, and performance-sensitive systems using Python, C++, Go, React, Docker, Redis, Postgres, and AWS.

## EDUCATION

**California State University, Sacramento**

Bachelor of Science, Computer Science

Expected December 2027

GPA: 3.47

## EXPERIENCE

**AI Engineer Extern, Pfizer** · *Extern*

June 2026 - Current

- Automated an OCR-backed document-processing pipeline to extract, classify, and route content from 1,000+ page pharmaceutical files.
- Improved document retrieval accuracy by **36%** by tuning document segmentation, indexing configuration, and model selection.
- Evaluated end-to-end structure of the document intelligence system, benchmarking OCR accuracy, RAG retrieval quality, and routing performance, identifying key areas of improvement for production deployment.
- Delivered a detailed technical report outlining model trade-offs, optimization strategies, and deployment recommendations.

**Transportation Security Officer** · *Trinity Technology Group*

July 2021 – June 2025

- Led coworkers and passengers in maintaining strict compliance with federal safety protocols to provide secure and efficient screening for thousands of passengers daily.

## PROJECTS

**GitHub Documentation RAG Service** · *Python, Qdrant, FastAPI, Docker, AWS*

- Engineered a pipeline for ingesting public GitHub repositories and answering documentation-based queries with citations.
- Implemented cross-encoder reranker with LLM-judged evaluator to ensure retrieval relevance for complex questions.
- Evaluated multiple indexing and ranking strategies, improving answer grounding to **92% faithfulness and 99% citation accuracy** on a 50-question evaluation suite.
- Containerized the FastAPI service with Docker and environment configurations for reproducible local and cloud deployment.
- Deployed the service behind a portfolio API with a web interface for interactive repository queries at [quwin.dev](https://quwin.dev).

**Nvidia CUDA GPU-Accelerated Postflop Poker Solver** · *C++, CUDA, FLTK*

- Built a custom No-Limit Hold'em postflop solver using Counterfactual Regret Minimization (CFR) across configurable game trees.
- Implemented CPU and GPU CFR execution with flattened device-side game data and CUDA kernels.
- Optimized solver memory architecture, **cutting memory usage by 60%** through batched hand-pair evaluation.
- Enabled near-linear vertical scaling for CFR iterations, reducing computation time by up to **27%** on entry-level consumer GPUs.

**Discord Community Utilities + Permissions Manager** · *Python, JavaScript, Postgres, Docker, Google Cloud*

- Developed and deployed a full-stack web platform with Discord verified utilities for community engagement to **100+ servers**.
- Engineered ingestion pipeline to prioritize important accounts, maintaining leaderboard freshness across millions of players.
- Reduced backend response time by 21% through connection pooling to support thousands of concurrent database calls.
- Integrated role-based access control for communities to create and manage user permissions linked to progression tracking.

**Distributed API Gateway + Rate Limiter** · *Go, Nginx, Redis, Docker, AWS*

- Architected a configurable production API gateway enforcing per-plan rate limits balanced across multiple instances.
- Implemented and benchmarked token bucket, sliding window, and fixed window algorithms under simulated burst load, reducing limit-bypass rate to zero across multi-instance deployment.
- Forwarded authenticated identity metadata upstream while stripping raw API keys from requests to preserve security.

## TECHNICAL SKILLS

**Programming:** C++, Python, Rust, Go, Java, JavaScript, TypeScript

**Frameworks:** LangChain, FastAPI, React.js, Next.js, Flask, Node.js

**Technologies:** AWS, Docker, Redis, Nginx, Postgres, Qdrant, HuggingFace, Git, Apache, Jira, Cursor, Claude Dev

**Libraries:** CUDA, PyTorch, TensorFlow, Pandas, Scikit-learn, JDBC, Tailwind, Plotly, Quart